PROJECT REPORT

ON

"AI-Based Underwater Monitoring System: Image Enhancement, Detection and Depth Mapping"

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE. IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE

BACHELOR OF ENGINEERING
IN
COMPUTER ENGINEERING

SUBMITTED BY

Miss. Pallavi R. Gawai	**Roll No. 4112**
Miss. Kajal A. Shelke	**Roll No. 4109**
Miss. Priyanka Gujale	**Roll No. 4150**
Miss. Dnyaneshwari Suke	**Roll No. 4152**

UNDER THE GUIDANCE OF

Prof. Archana Kotakar

DEPARTMENT OF COMPUTER ENGINEERING

JSPM's Jayawantrao Sawant College of Engineering, Pune

Hadapsar, Tal-Haveli, Pin- 411028, India

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JAYAWANT SHIKSHAN PRASARAK MANDAL's
Jayawantrao Sawant College of Engineering
(Approved by AICTE, New Delhi, Govt of Maharashtra and Affiliated to University of Pune)
Id.No. : PU/PN/Engg./199/(2004)
S. No.58, Handewadi Road, Hadapsar, Pune - 411028
Ph. :7498934414/ 8669605807
Email : principal@jspmjscoe.edu.in
Website : www.jspmjscoe.edu.in

Prof.Dr.T.J.Sawant
D.E.E., B.E.(Electrical), MISTE,Ph.D
FOUNDER SECRETARY

Prof.Dr. Pradeep Anandrao Patil
Ph.D, (Mechanical) : HVAC & Heat Transfer
BE (Mechanical)
ME (Mechanical) Heat Power Engineering
Principal

DEPARTMENT OF COMPUTER ENGINEERING

JSPM's Jayawantrao Sawant College of Engineering, Pune

Hadapsar, Tal-Haveli, Pin- 411028, India

CERTIFICATE

This is to certify that the Project report entitled

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Submitted by

Miss. Pallavi R. Gawai	**Roll No. 4112**
Miss. Kajal A. Shelke	**Roll No. 4109**
Miss. Priyanka Gujale	**Roll No. 4150**
Miss. Dnyaneshwari Suke	**Roll No. 4152**

This is a bonafide work carried out under the supervision of Prof. Archana Kotakar and it is submitted towards the partial fulfillment of the requirement of Savitribai Phule Pune University, Pune for the award of the degree of Bachelor of Engineering in Computer Engineering, at JSPM'S Jayawantrao Sawant College of Engineering, Pune.

Prof. Archana Kotakar
Internal Guide
Department of Computer Engineering

External Examiner

Prof.A.S. Dake
Project Coordinator
Department of Computer Engineering

Dr. P. D. Lambhate
Head of Department
Department of Computer Engineering

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Miss. Pallavi R. Gawai **Roll No.4112**

Miss. Kajal A. Shelke **Roll No.4109**

Miss. Priyanka Gujale **Roll No. 4150**

Miss. Dnyaneshwari Suke **Roll No. 4152**

(B.E. Computer Enginering)

JSCOE, Pune

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1. Introduction

In recent years, environmental pollution in water bodies has become a critical global concern, particularly in regions with limited monitoring infrastructure and growing urbanization. Human activities such as industrial discharge, dumping of plastics, and improper waste management have caused severe degradation of underwater ecosystems. Traditional methods for monitoring and cleaning these areas rely heavily on manual inspection, divers, or simple sensors—approaches that are often inefficient, dangerous, and lack real-time data processing capabilities. The growing need for smart and automated monitoring systems has driven the development of AI-based underwater monitoring solutions, which use artificial intelligence, deep learning, and computer vision to detect and analyze underwater waste effectively.

This project proposes an intelligent and automated system capable of performing image enhancement, object detection, and depth mapping to understand the environmental status of underwater regions. By utilizing high-performance models such as YOLOv8 for object detection and MiDaS v3 for depth estimation, the system can identify waste objects and their spatial distribution in real time.

1.1.1 Demand Forecasting

In the context of environmental management, demand forecasting refers to predicting the future state or intensity of pollution and resource requirements for cleaning and maintenance. The proposed system applies this principle by forecasting the pollution severity in different underwater zones based on real-time image data, depth estimation, and historical patterns. Using advanced analytical methods, the system predicts how pollution levels may change over time and which areas are likely to become critical in the near future. This predictive insight helps authorities allocate manpower, cleaning robots, and other resources efficiently, ensuring timely intervention.

Through AI-based forecasting, the system bridges the gap between observation and preventive action. For example, if continuous monitoring shows an increasing density of plastic or metallic waste in a certain zone, the system can predict that the area will shift from a "moderate" to a "critical" pollution level soon.

1.1.2 Challenges of Infrastructure

- Capturing clear underwater images is difficult due to low visibility, murky water, and light distortion.
- Enhancing and processing large volumes of video and image data in real time requires high computational power.
- Ensuring accurate detection of garbage using AI models in complex underwater conditions is challenging.
- Reliable data transfer and system operation in remote or deep water areas can be limited by connectivity.
- Adapting AI models like YOLOv8 and MiDaS v3 for underwater environments requires careful optimization and training.

1.3 Machine Learning for Prediction

Machine learning plays a central role in the proposed AI-based underwater monitoring system by enabling intelligent detection, classification, and prediction of pollution severity in real time. Traditional methods of monitoring water bodies rely on manual observation, which is inefficient, subjective, and often unsafe. By integrating machine learning models, the system can analyze underwater images and videos automatically, identify waste materials, estimate their density, and classify polluted zones accurately. This predictive capability allows environmental authorities to prioritize cleanup operations and allocate resources more effectively.

The system utilizes YOLOv8, a state-of-the-art deep learning model, for object detection. YOLOv8 can detect multiple types of waste simultaneously, even in challenging underwater conditions, ensuring high accuracy and speed. For depth estimation, the system employs MiDaS v3, which converts 2D images into 3D depth maps. This depth information, combined with detected objects, allows the system to determine the severity and spread of pollution across different zones.

1.4 Motivation

Water pollution is one of the most pressing environmental challenges today, significantly affecting rivers, lakes, ponds, and other aquatic ecosystems. Industrial discharge, plastic waste, metallic debris, and organic pollutants have increasingly contaminated these water bodies, posing risks to biodiversity, human health, and overall ecological balance. Traditional monitoring and cleanup methods rely heavily on manual inspections and periodic surveys, which are often inefficient, time-consuming, and dangerous, especially in areas with poor visibility, strong currents, or deep water. The lack of real-time data and automated analysis has made it difficult for authorities to take timely and informed action against underwater pollution.

The motivation for this project arises from the urgent need to develop an intelligent, automated, and safe system that can monitor underwater environments continuously and accurately. By integrating artificial intelligence, deep learning, and computer vision techniques, the system enhances underwater images, detects and classifies different types of waste, and maps the depth and density of polluted zones. This capability not only ensures precise identification of polluted areas but also enables predictive analysis, allowing authorities to forecast potential hotspots and allocate cleanup resources efficiently. Such data-driven insights are crucial for improving the effectiveness, safety, and planning of environmental interventions.

1.5 Objectives

1. Underwater Image Enhancement:

To improve the quality of underwater images and videos by removing noise, correcting colors, and enhancing contrast using advanced OpenCV-based image processing techniques. This ensures better visibility in murky or distorted underwater conditions, which is essential for accurate detection and analysis.

2. Garbage Detection and Classification:

To accurately detect and classify different types of underwater waste, such as plastics, cans, and organic materials, using the YOLOv8 deep learning

3. Depth Mapping and Zone Analysis:

To estimate the depth of underwater areas using the MiDaS v3 model and calculate the density of garbage. By combining depth information with detected objects, the system can classify zones as clean, moderate, or critical, helping in prioritizing cleanup operations efficiently.

4. Predictive Analysis Using Machine Learning:

To apply K-Means clustering and Decision Tree algorithms for analyzing and predicting pollution severity in different zones. This predictive capability allows authorities to forecast potential hotspots and plan interventions proactively.

5. Dashboard Visualization for Actionable Insights:

To develop an interactive dashboard that displays enhanced images, detected garbage, depth maps, and classified zones. This visualization helps environmental authorities and maintenance workers make data-driven decisions for efficient resource allocation and timely cleanup operations.

2. Literature Survey

Chaudhary et al proposed an early deep learning approach to enhance underwater images using convolutional neural networks, addressing issues such as haze, color distortion, and poor visibility. Their method significantly improved the visibility and quality of underwater images, forming the foundation for later research in underwater enhancement and detection. [1]

Islam et al. introduced a real-time underwater image enhancement model to improve visual perception and clarity in turbid water conditions. The approach used deep learning-based techniques to restore color balance and reduce scattering effects, resulting in enhanced underwater imagery suitable for environmental monitoring. [2]

Zhao et al.developed a real-time underwater garbage detection algorithm using the YOLOv3 framework. This model effectively detected underwater waste objects and achieved better accuracy for visible targets, although its performance decreased when detecting small or partially hidden objects in low-visibility waters. . [3]

Song et al. improved underwater object detection by integrating attention mechanisms into the YOLO architecture. This enhancement resulted in higher detection accuracy and better recognition in complex underwater environments with varying lighting and visibility conditions. [4]

Li et al. presented a medium transmission-guided enhancement technique that used multiple color spaces to enhance underwater images and improve object detection performance. The method achieved natural color restoration and improved detection accuracy but did not integrate depth estimation modules. [5]

Islam et al. proposed a semantic segmentation-based approach for identifying marine debris in underwater videos. The model demonstrated efficient detection of overlapping and partially visible waste materials, providing better performance in debris classification and monetring. [6]

Sr. No.	Title	Year	Author(s)	Remark (What they did)	Gap Analysis
1	Underwater Object Detection with Deep Learning (EUVIP Conference)	2023	D. L. Biagi, D. D. Blo, Truzzisi, F. Narducci, and L. Veltri	Focused on object detection using CNNs and YOLO for underwater environments. Achieved accurate detection in clear conditions.	Did not include image enhancement or depth estimation, resulting in reduced accuracy in turbid or low-visibility waters.
2	Underwater Target Detection Using Deep Learning Methodologies: Challenges, Applications, and Future Evolution (IEEE Access)	2023	S. Khunteta, V. Kaushik, N. Kumar, and D. K. Sharma	Surveyed deep learning approaches for underwater detection and discussed challenges and applications.	Purely a review; lacks implementation of an integrated system combining enhancement, detection, and depth mapping.
3	Deep Learning Innovations for Underwater Waste Detection: An In-depth Analysis (arXiv Preprint)	2024	J. S. Walia and P. L. K.	Applied deep learning models to detect underwater waste objects for pollution monitoring.	Limited to waste detection only; did not address image enhancement or depth estimation.
4	Deep Learning Innovations for Underwater Waste Detection: Extended Study (IEEE Access)	2025	J. S. Walia, K. Haridass, and P. L. K., VIT Chennai	High computational cost; lacks comparison raw detection , no full integration enhancement, detection, depth visualization.	High computational cost; lacks comparison with raw detection and no full integration of enhancement, detection, and depth visualization.

2.1 Reference Paper Comparison Table

3.Problem Statement

Water pollution in rivers, ponds, and lakes has become a serious environmental issue due to the accumulation of plastics, cans, and organic waste. Traditional monitoring and cleaning methods are often inefficient, unsafe, and time-consuming, especially in areas with poor visibility or deep water. Manual inspections fail to provide real-time insights, and existing solutions do not offer a complete understanding of pollution severity, making it difficult for authorities to plan effective cleanup operations.

To address this challenge, our project develops an AI-based underwater monitoring system that integrates image enhancement, object detection, depth mapping, and zone classification. Using YOLOv8 for waste detection, MiDaS v3 for depth estimation, and machine learning techniques like K-Means clustering and Decision Trees, the system can classify zones as clean, moderate, or critical.

3.1 Project Scope

- The scope of this project defines the boundaries and applicability of the AI-based underwater monitoring system:
- The system can capture and process underwater images and videos for monitoring water bodies such as ponds, lakes, and rivers.
- It provides real-time image enhancement to improve clarity in murky or low-light conditions.
- The system detects and classifies various types of garbage, including plastics, cans, and organic waste, using YOLOv8.
- It estimates depth and density of garbage zones with MiDaS v3 and classifies areas as clean, moderate, or critical using K-Means and Decision Tree algorithms.
- The project provides a dashboard for visualization, enabling authorities to view pollution severity, depth maps, and actionable insights.
- The system focuses on automation, safety, and efficiency, reducing the need for manual inspections and allowing proactive environmental management.

3.2 Mathematical Model

1. Image Enhancement (OpenCV Based)

Let
I(x, y) = input underwater image (low visibility)

The enhanced image Ie(x, y) can be represented as:

$$Ie(x, y) = f(F(I(x, y)))$$

where

- F(I): filtering operation such as Gaussian or Bilateral filter for denoising
- f(): contrast or brightness adjustment function such as CLAHE or Histogram Equalization

Example operations:

$$I_{denoise} = I * G_\sigma$$
$$I_{enhanced} = CLAHE(I_{denoise})$$

This process improves image color, contrast, and visibility before object detection.

2. Object Detection (YOLOv7 Based)

Let the enhanced image be I_e.
YOLOv7 divides I_e into an $S \times S$ grid, and each cell predicts bounding boxes and class probabilities.

Equations used:

$$P_{obj} = \sigma(C)$$
$$B = (x, y, w, h)$$
$$Confidence = P_{obj} \times P_{class}$$

where

- P_{obj}: objectness score
- P_{class}: class probability (for example fish, coral, debris)
- B: bounding-box coordinates
- σ: sigmoid activation

The total loss minimized by YOLOv7 is:

$$L = L_{box} + L_{obj} + L_{class}$$

This represents the sum of box regression loss, objectness loss, and classification loss.

3. Depth Mapping (MiDaS v3)

MiDaS v3 performs monocular depth estimation from a single enhanced image.

Given the image I_e, the depth map D is obtained as:

$$D = M_{MiDaS}(I_e)$$

where $D(x, y)$ gives the relative depth for each pixel.

Normalization to make depth values comparable:

$$D_n(x, y) = (D(x, y) - D_{min}) / (D_{max} - D_{min})$$

This ensures consistent depth representation for different images.

4. Zone Analysis (K-Means + Decision Tree)

The final step involves clustering and classification to determine underwater zones such as Safe, Critical, or Debris-Rich.

Feature vectors:

$$\mathbf{X = \{x1, x2, \ldots, xn\}}$$

represent color, texture, and depth features.

K-Means minimizes:

$$\mathbf{J = \Sigma(i = 1 \text{ to } k)\ \Sigma(xj \in Ci)\ ||xj - \mu i||^2}$$

where μi is the centroid of cluster Ci.

A Decision-Tree classifier is then applied on the clustered data to label each zone accordingly.

Overall System Flow

Input Image → Image Enhancement → YOLOv7 Detection → MiDaS Depth Estimation → K-Means + Decision Tree → Zone Classification

4.System Design And Implementation

4.1 Diagrams

4.1.1 Flow Chart

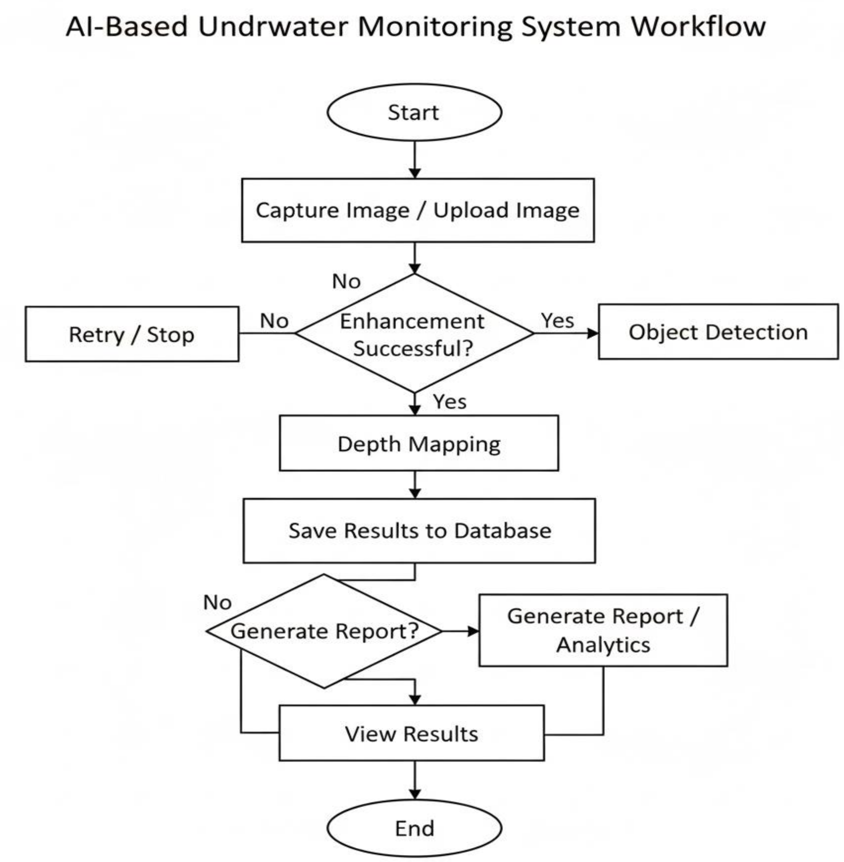


Fig 4.1.1 flow chart

- The system starts by capturing or uploading an underwater image for analysis.
- The input image is enhanced using AI-based algorithms to improve visibility, contrast, and color balance.
- If the enhancement is unsuccessful, the system allows the user to retry or stop the process.
- Once enhanced successfully, the system performs object detection to identify marine life, debris, or other underwater objects.
- Depth mapping is then applied to estimate the distance and spatial position of the detected objects.
- The user can choose to generate analytical reports or directly view the results.
- Finally, the system displays the **processed outputs and concludes** the workflow.

4.1.2 UML Diagram

4.1.2.1 Sequence Diagram

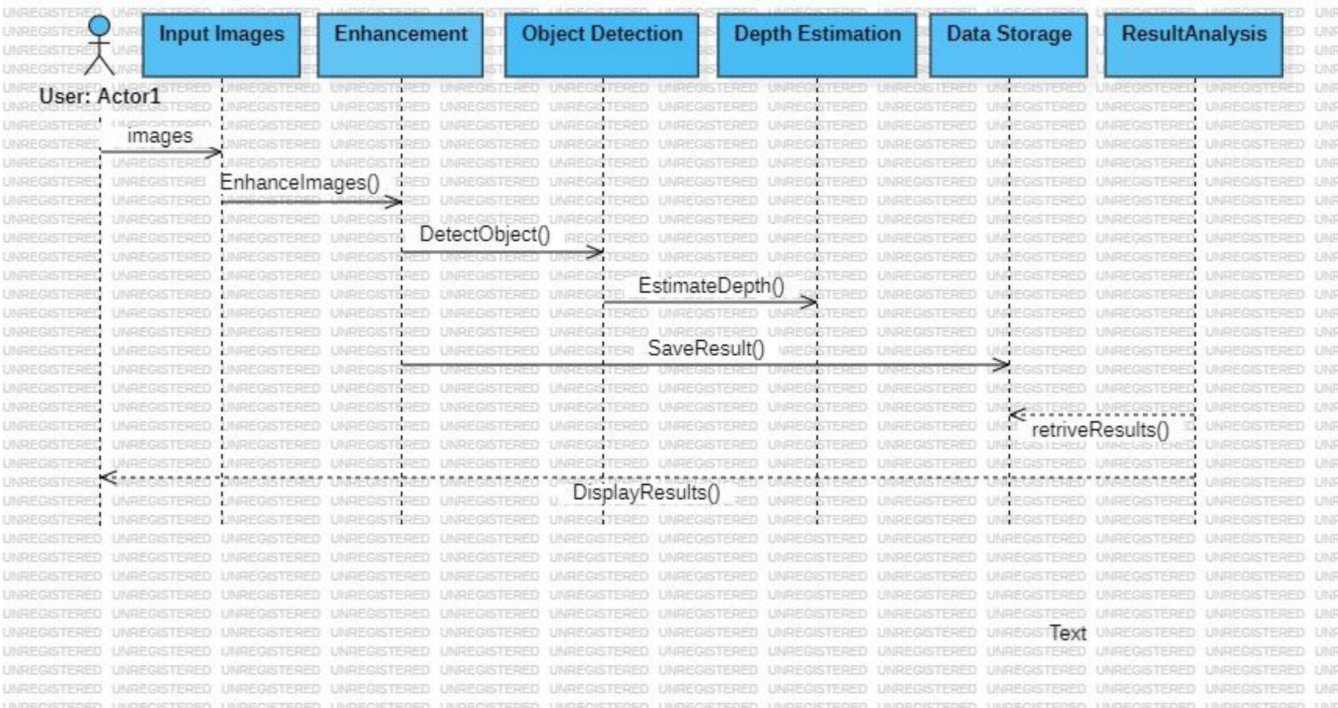


Fig 4.1.2 Sequence Diagram

The diagram represents the workflow of the AI-Based Underwater Monitoring System, showing the sequence of steps involved in processing underwater images for enhancement, detection, and analysis.

- InputImages represents the raw underwater images captured or uploaded by the user.
- Enhancement improves image clarity, color balance, and visibility using AI-based image enhancement techniques.
- ObjectDetection identifies underwater objects such as marine life, corals, or debris using detection algorithms.
- DepthEstimation calculates the relative distance and spatial depth of detected objects using AI models.
- DataStorage saves the processed and analyzed results in a database for future access and reporting.
- ResultAnalysis retrieves and displays the enhanced images, detection results, and depth data for visualization and interpretation.

4.1.2.2 Class Diagram

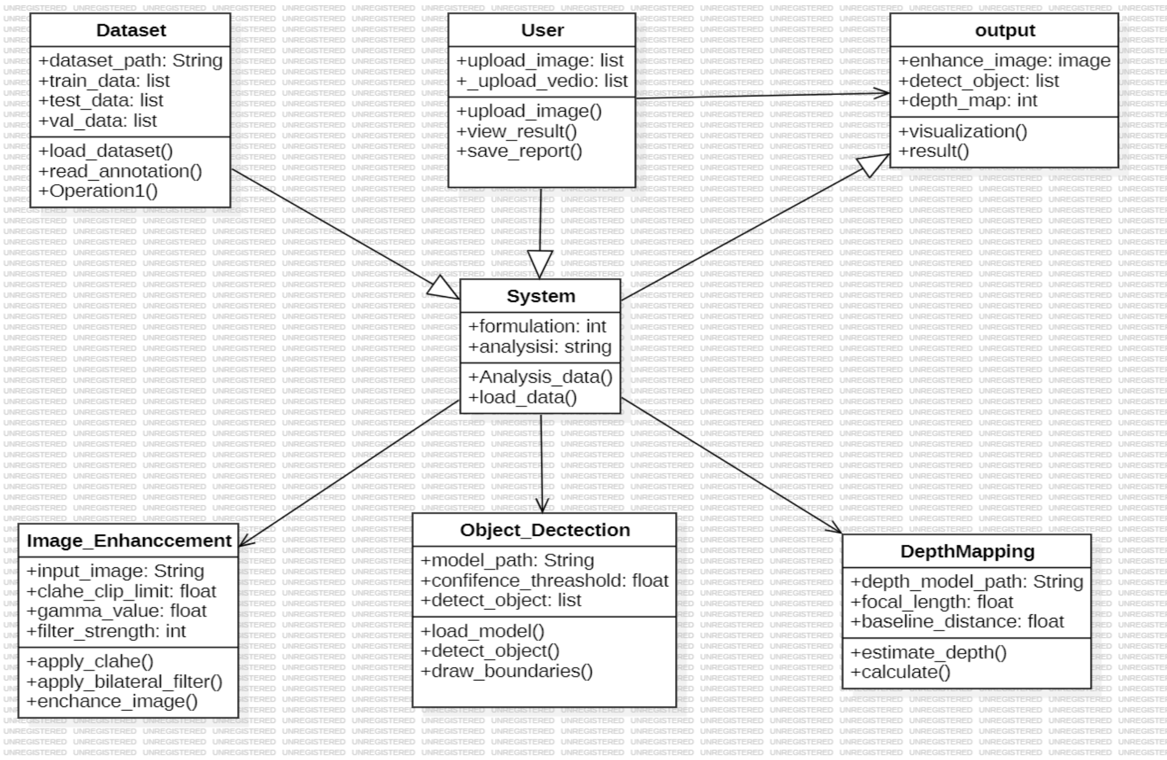


Fig 4.1.3 Class diagram

The diagram represents the Class Diagram of the AI-Based Underwater Monitoring System, showing the main components, their attributes, and interactions for image processing and analysis.

- Dataset handles loading & reading of training, testing, and validation datasets used for model operations.
- User represents the end-user who uploads images or videos, views processed results, and saves analytical reports.
- System coordinates the overall workflow, manages data analysis, and controls communication between modules.
- Image Enhancement improves underwater image visibility using techniques like CLAHE, gamma correction, and bilateral filtering.
- Object Detection identifies underwater objects by loading AI models and drawing detection boundaries around recognized entities.
- Depth Mapping estimates the distance and depth of detected objects using parameters like focal length and baseline distance.

- Output stores the enhanced images, detected objects, and depth maps, providing visualization and result generation functionalities

4.1.2.3 Use Case Diagram

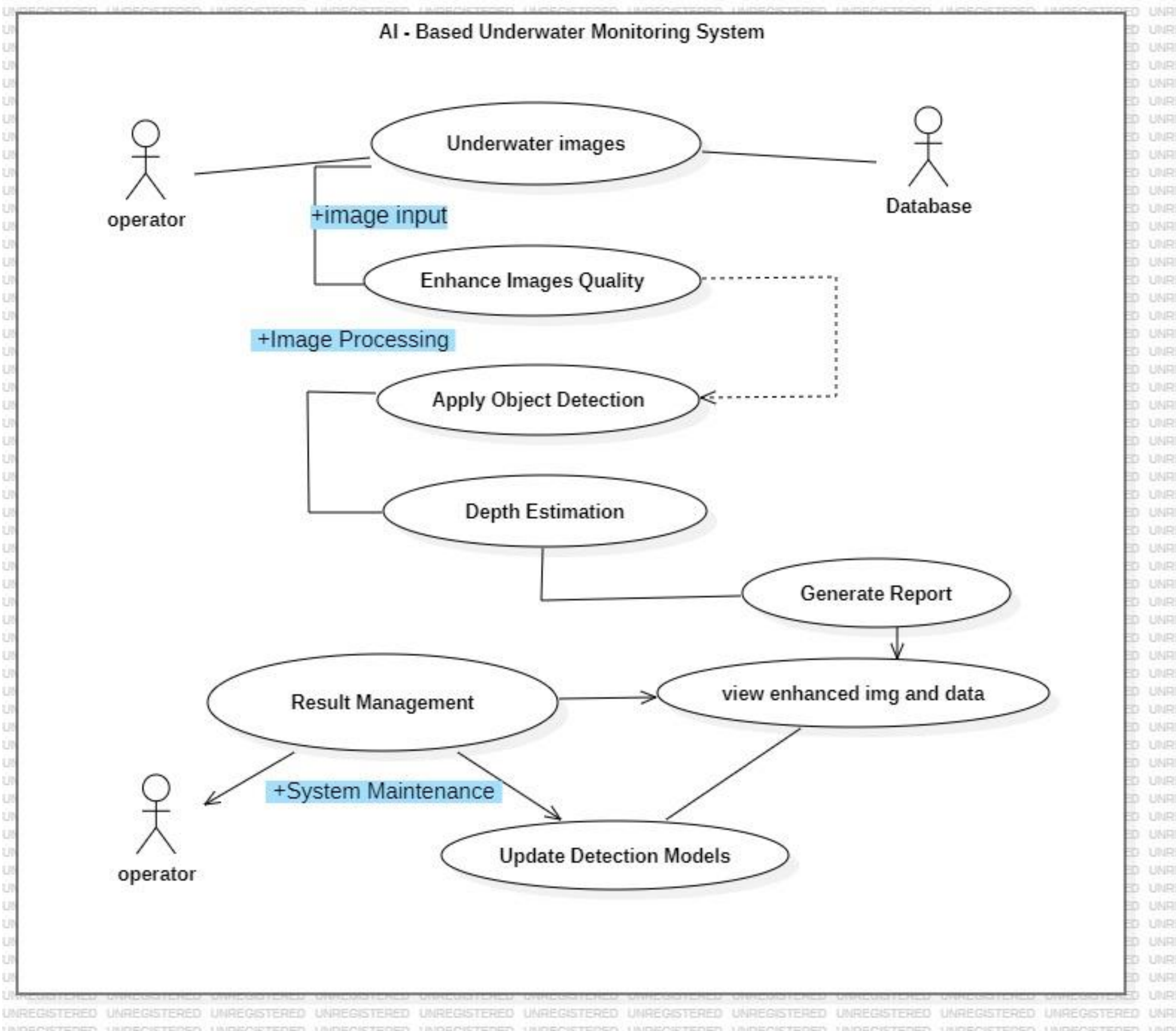


Fig 4.1.4 use case diagram

The diagram represents the Use Case Diagram of the AI-Based Underwater Monitoring System, illustrating the interaction between the operator, the database, and the system's core functionalities.

- Underwater Images: The operator provides underwater images as input to the system for processing.

- Enhance Images Quality: The system improves the visual clarity of underwater images by reducing noise and correcting colors.
- Apply Object Detection: AI algorithms detect and identify underwater objects such as marine life, corals, or debris.
- Depth Estimation: The system calculates the distance and depth of detected objects to understand spatial relationships.
- Generate Report: Based on analysis, the system generates a detailed report including detected objects and depth information.
- Result Management: The operator can view enhanced images, detection results, and stored data.
- Update Detection Models: The system is regularly updated to maintain accuracy and efficiency in detection and analysis.
- System Maintenance: The operator manages updates and ensures smooth system functionality

4.1.2.4 Activity Diagram

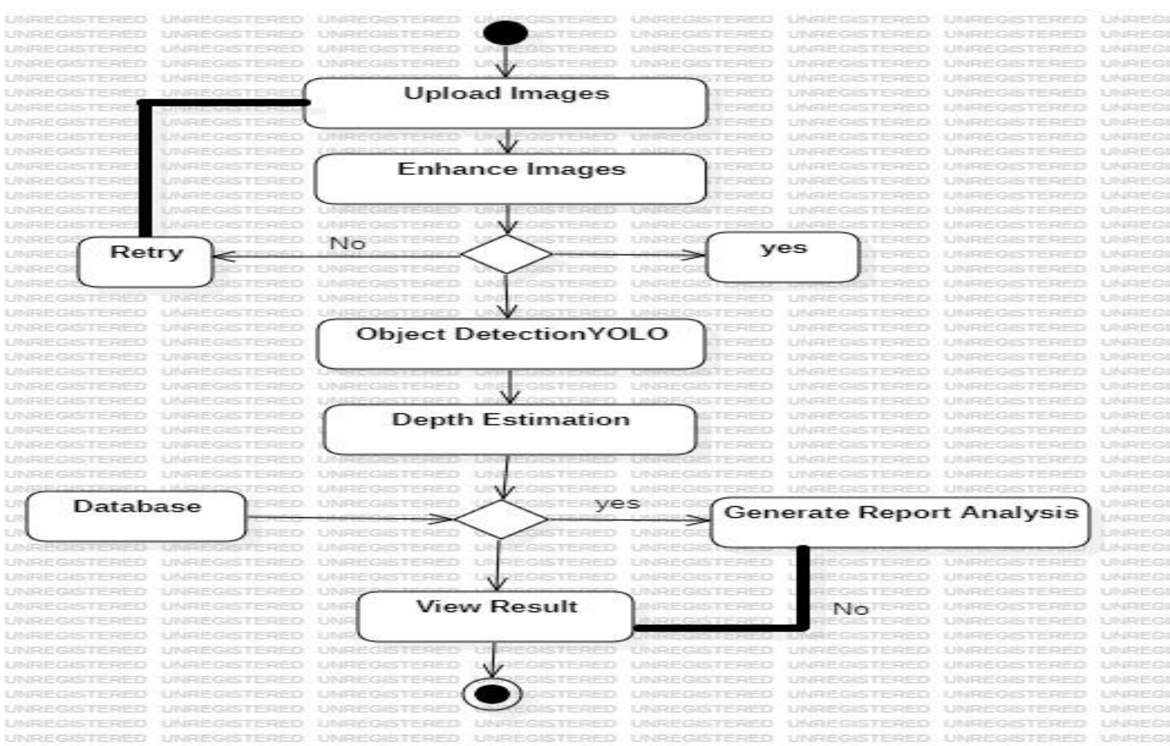


Fig 4.1.5 Activity diagram

- The process begins with uploading underwater images into the system.

- The system performs image enhancement to improve visibility and quality.
- If enhancement succeeds, it continues to YOLO-based object detection and depth estimation.
- The analyzed data and results are stored in the database.
- A report or analysis is generated based on user selection.
- Finally, the user can view the enhanced images and final results.

4.2 System Features

1. Real-Time Image Enhancement

The system enhances underwater images and video frames in real time using OpenCV-based algorithms. It removes noise, adjusts contrast, and corrects color distortion to improve visibility in murky or low-light conditions.

2. Garbage Detection and Classification

The system uses YOLOv8, a state-of-the-art deep learning model, to detect and classify underwater waste such as plastics, cans, and organic material. Detected objects are accurately localized in the enhanced images, even in challenging underwater conditions.

3. Depth Estimation

Depth mapping of detected objects is performed using MiDaS v3, which generates a 3D representation from 2D images. This helps in calculating the depth and density of garbage in different underwater zones.

4. Zone Classification

The system applies K-Means clustering and Decision Tree algorithms to classify water zones as clean, moderate, or critical. This classification is based on the density and depth of detected garbage, enabling an organized understanding of pollution levels.

5. Dashboard Visualization

All processed data is displayed on a user-friendly dashboard, showing enhanced images, detected objects, depth maps, and classified zones. This visualization provides actionable insights for environmental authorities and workers.

4.3 System Implementation Constraints

1. Hardware Limitations: The system relies on cameras and computational hardware for real-time processing. High-performance GPUs or processing units are required for YOLOv8 and MiDaS v3, which may limit deployment in resource-constrained environments.

2. Water Conditions: Murky water, low lighting, suspended particles, or turbidity can affect image quality, making detection and depth estimation less accurate. Extreme environmental conditions may require additional enhancement techniques.

3. Real-Time Processing: Processing large volumes of video frames in real time requires significant computational power. Delays may occur if the system hardware cannot handle the high data throughput efficiently.

4. Connectivity Constraints: Real-time monitoring may be affected in remote or deep water areas with limited network connectivity, which can impact dashboard updates and data transfer.

5. Scalability: Monitoring large water bodies or multiple zones simultaneously may require scaling the system hardware and AI models, which could increase deployment cost and complexity.

5.Software Requirement and Specification

5.1 Introduction

The Software Requirements and Specifications define the essential tools, frameworks, and algorithms needed to implement the AI-based underwater monitoring system efficiently. This system integrates image processing, deep learning, and machine learning techniques to provide real-time monitoring, garbage detection, depth estimation, and zone classification in water bodies such as rivers, lakes, and ponds.

5.2 Algorithm Selection

Image Enhancement, Garbage Detection, and Depth Estimation

This algorithm focuses on preprocessing and analyzing underwater images to detect and estimate the depth of garbage.

Algorithm 1: Image Enhancement (OpenCV):

Apply noise removal, contrast adjustment, and color correction to underwater frames.

Output: Enhanced images () suitable for detection and depth estimation.

Algorithm 2. Garbage Detection (YOLOv8):

Input: Enhanced images ()

Detect and classify underwater waste such as plastics, cans, and organic materials.

Output: Detected objects () with bounding boxes and class labels.

Algorithm 3. Depth Estimation (MiDaS v3):

Input: Enhanced images ()

Generate depth maps for detected objects to calculate distance and density.

Output: Depth values () corresponding to each object.

Algorithm 4: Zone Classification and Analysis

This algorithm focuses on clustering and classifying polluted zones based on object detection and depth data.

1. K-Means Clustering:

Input: Detected objects () and depth values ()

Cluster areas based on garbage density and depth to identify potential pollution zones.

Output: Clustered zones () representing areas with similar pollution levels.

2. Decision Tree Classification:

Input: Clustered zones ()

Classify zones as clean, moderate, or critical based on object density and depth criteria.

Output: Final classified zones () for actionable insights.

3. Dashboard Visualization:

Display enhanced images, detected objects, depth maps, and classified zones. Enable real-time monitoring and decision-making for environmental authorities.

5.3 Hardware Requirements

- The AI-based underwater monitoring system requires the following hardware to ensure smooth operation and real-time processing:
- Processor: Intel Core i5 or equivalent (8th generation or higher recommended)
- RAM: 8 GB minimum (16 GB recommended for real-time video processing)
- Storage: 500 GB HDD/SSD (for storing images, videos, and processed data)
- Display: 14-inch or higher monitor for dashboard visualization

5.4 Software Requirements

- Operating System: Windows 10/11 or Linux Ubuntu 20.04+
- Programming Language: Python 3.8+
- Libraries & Frameworks:OpenCV (for image enhancement)
- PyTorch / TensorFlow (for YOLOv8 and MiDaS v3 models)
- Scikit-learn (for K-Means and Decision Tree algorithms)
- Matplotlib / Plotly / Dash (for dashboard visualization)
- IDE / Development Environment: Visual Studio Code, Jupyter Notebook, or PyCharm

5.5 Software Quality Attributes

1. Performance:

The system performs real-time image enhancement, garbage detection, and depth estimation efficiently, even with high-resolution video inputs.

Optimized algorithms and GPU acceleration ensure minimal processing delay.

2. Accuracy:

YOLOv8 provides high-precision garbage detection, and MiDaS v3 ensures accurate depth estimation.

K-Means clustering and Decision Tree classification enable correct zone categorization, reducing errors in pollution assessment.

3. Reliability:

The system consistently processes underwater data under various conditions, including low light and moderate turbidity.

Robust error handling ensures continuous operation without crashes.

4. Scalability:

Modular design allows the system to handle larger water bodies or multiple zones by adding additional cameras or computing resources.

5. Usability:

The dashboard provides a user-friendly interface, making it easy for environmental authorities to monitor zones, view depth maps, and analyze pollution levels.

5.6. Analysis Model

The Analysis Model defines how the AI-based underwater monitoring system processes input data, analyzes it, and produces actionable insights. It focuses on breaking down the problem, identifying system components, and showing the flow of data through the system.

1. Input Module:

Receives underwater images or video frames.

Prepares the data for processing without manual intervention.

2. Processing Module:

Image Enhancement: Uses OpenCV algorithms to improve visibility and clarity.

Garbage Detection: YOLOv8 detects and classifies objects in enhanced images.

Depth Estimation: MiDaS v3 generates depth maps to determine the distance and density of objects.

3. Analysis Module:

K-Means Clustering: Groups detected objects into zones based on density and depth.

Decision Tree Classification: Assigns each zone a pollution severity level (clean, moderate, critical).

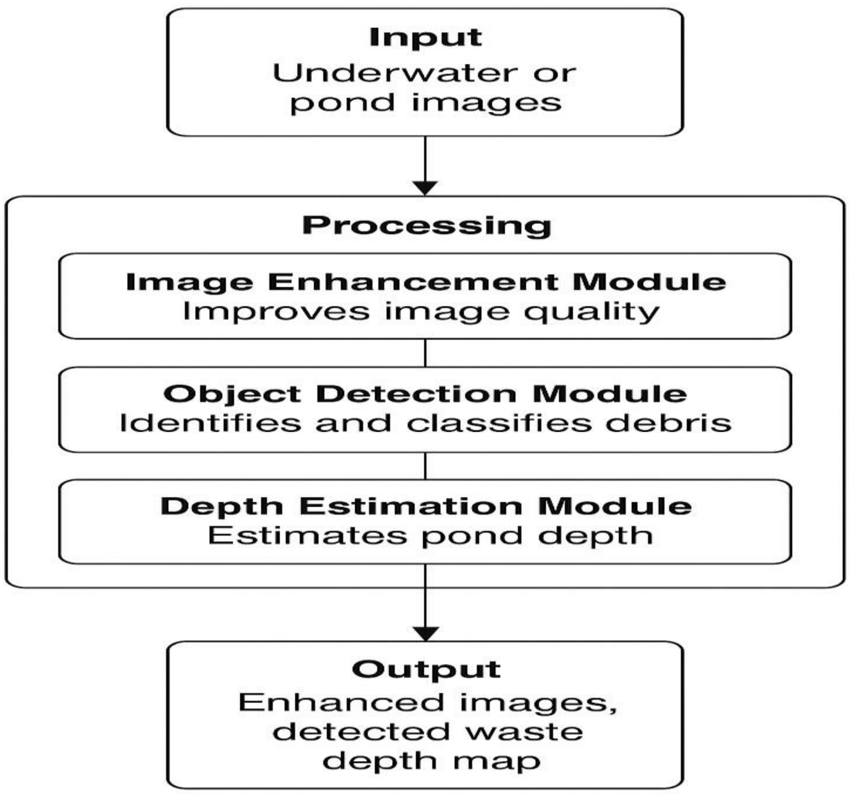


Figure 1: Analysis Model

6. Project Plan

The project plan for the AI-based underwater monitoring system is organized to ensure systematic development, integration, and deployment. The key points are:

1. Requirement Analysis:

Identify the objectives of the system, including image enhancement, garbage detection, depth mapping, and zone classification.

Determine hardware and software requirements, constraints, and system quality attributes.

2. System Design:

Design the overall system architecture, including modules for image processing, AI-based detection, depth estimation, and dashboard visualization.

Define data flow, algorithms, and modular interactions to ensure seamless integration.

3. Module Implementation:

Develop the Image Enhancement module using OpenCV to improve underwater visibility.

Implement Garbage Detection using YOLOv8 for object localization and classification.

Integrate Depth Estimation using MiDaS v3 to calculate the depth and density of objects.

Apply K-Means clustering and Decision Tree algorithms for zone classification.

Phase	**Description**	**Duration**	**Deliverables**
Phase 1: Requirement Analysis	Identify project needs, including software, hardware, and dataset for underwater image enhancement and waste detection. Define system objectives and success criteria.	2 Weeks	Requirement Specification Document

Phase	Description	Duration	Deliverables
Phase 2: System Design	Prepare architecture diagrams and workflow models showing how enhancement, detection, and depth mapping will work together.	2 Weeks	System Design and Architecture Model
Phase 3: Module Implementation	Develop modules for image enhancement (OpenCV), waste detection (YOLOv8), and depth estimation (MiDaS). Integrate all with a dashboard interface.	4 Weeks	Functional Modules and Source Code
Phase 4: Integration and Testing	Combine all modules into a single system. Test for accuracy, speed, and stability using underwater sample datasets.	3 Weeks	Integrated System and Test Report
Phase 5: Result and Analysis	Evaluate performance, collect outputs, generate visual results, and analyze detection accuracy and depth mapping.	2 Weeks	Output Screenshots, Result Graphs
Phase 6: Documentation and Presentation	Prepare final report, user manual, and presentation slides. Include results, system features, and future improvements.	2 Weeks	Final Report and Presentation Slides

7. Result

7.1 Expected Result

The proposed **AI-Based Pond Monitoring and Cleaning Assistance System** aims to deliver an intelligent, real-time solution for analyzing pond environments using advanced image processing and machine learning. The system enhances pond images, detects debris or waste materials, and estimates depth variations to support effective and safe cleaning by workers.

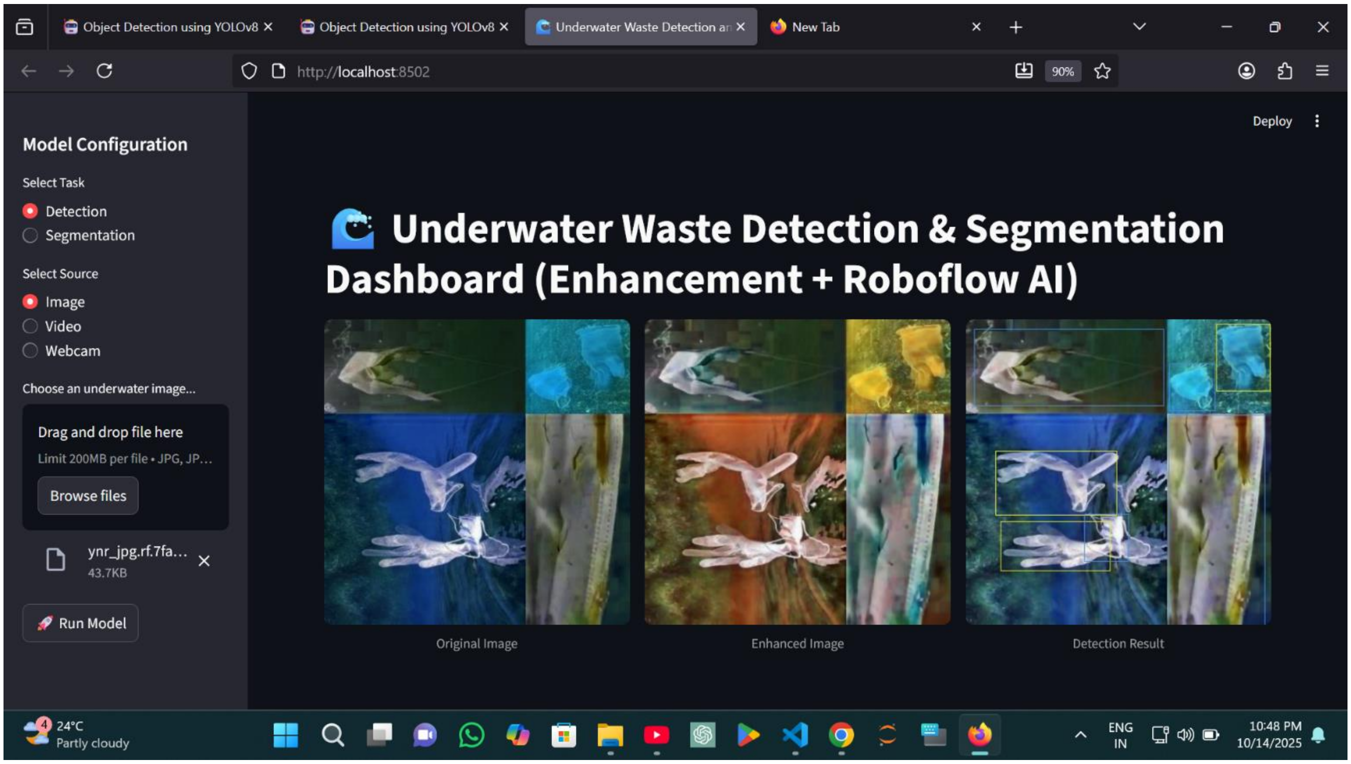
The project combines **image enhancement**, **object detection (YOLO)**, and **depth estimation (MiDaS)** models to provide accurate, easy-to-understand results for real-world pond maintenance.

Key expected outcomes include:

1. **Enhanced Pond Image Quality:**
   The system significantly improves the clarity, brightness, and contrast of pond images captured in turbid or unclear conditions, helping in accurate monitoring.
2. **Accurate Debris and Waste Detection:**
   Using the YOLO model, floating waste, algae, and plastic debris are automatically identified and highlighted, helping workers locate polluted areas easily.
3. **Depth Estimation and Mapping:**
   The MiDaS model generates detailed depth maps of the pond bed, allowing the team to identify shallow or deep regions, reducing the risk of over-cleaning or equipment damage.
4. **Automated Cleaning Zone Recommendation:**
   Based on debris concentration and depth variation, the system categorizes pond zones into **priority**, **moderate**, and **clean** areas for targeted cleaning operations.
5. **User-Friendly Visualization Interface:**
   An interactive graphical interface displays the original image, enhanced version, detected debris, and depth map for easy comparison and analysis by workers.
6. **Improved Worker Efficiency and Safety:**
   By providing actionable visual information, the system minimizes manual inspection and reduces time, cost, and physical effort in pond cleaning tasks.
7. **Scalability and Adaptability:**
   The model can be adapted for various pond sizes, water conditions, and environments by retraining on new image datasets or integrating IoT-based sensors in the future.

Overall, the proposed system delivers **a robust, AI-powered tool** for sustainable pond management — improving water quality, optimizing cleaning schedules, and ensuring a safer working process for cleaning staff.

7.2 Snapshots



Model Dashboard Interface

The main interface of the Underwater Waste Detection Dashboard allows users to select tasks (Detection) upload underwater images, and run AI models for analysis.

Image Enhancement Result

This module enhances low-visibility underwater images using deep learning algorithms.

- *Left:* Original Image
- *Middle:* Enhanced Image
  The enhanced output shows improved brightness, contrast, and clearer object boundaries.

Detection Output

After enhancement, the YOLOv8 model detects waste materials such as plastic and debris.

- *Right:* Detection Result
  Bounding boxes indicate accurately detected waste regions, making the system efficient for underwater monitoring and research.

8 Advantages, Disadvantages and Application

Advantages

1. Improved Image Quality:

Enhances low-quality, blurry, or noisy images using GANs and CLAHE, making details clearer.

2. Accurate Object Detection:

Detects and classifies multiple objects in images, even under challenging conditions like low light or underwater environments.

3. Depth Estimation:

Provides depth maps, enabling spatial understanding of the scene and relative distances between objects.

4. Automation:

Reduces manual effort in image enhancement and analysis, allowing automatic detection and depth estimation.

5. Scalability:

Can handle large datasets, higher-resolution images, or additional features like segmentation in future versions.

Disadvantages

1. High Computational Requirement:

Training GANs and deep learning models requires powerful GPUs and high memory, which may not be available on all systems.

2. Data Dependency:

Model accuracy depends on the quality and quantity of training datasets. Poor or insufficient data can reduce performance.

3. Complexity in Integration:

Combining image enhancement, object detection, and depth learning modules can be challenging and may require careful tuning.

4. Training Time:

Deep learning models, especially GANs, take significant time to train, even with GPU acceleration.

5. Overfitting Risk:

If the model is trained on a limited dataset, it may perform poorly on unseen images.

Applications

1. Underwater Image Analysis:

Enhances visibility in underwater photography or research, and helps in detecting marine life or objects.

2. Autonomous Vehicles:

Supports object detection and depth perception for navigation in complex environments.

3. Robotics and Drones:

Assists robots and drones in recognizing objects and understanding spatial relationships.

4. Medical Imaging:

Improves image clarity for diagnostics and helps detect anomalies in scans like X-rays or MRIs.

5. Surveillance and Security:

Enhances CCTV footage for better detection of objects, intruders, or suspicious activities.

6. Satellite and Aerial Imaging:

Enhances low-resolution satellite images and detects objects or land features accurately.

9.Conclusion:

The project "AI-Based Image Enhancement, Object Detection, and Depth Learning Using Deep Learning" successfully demonstrates the integration of advanced artificial intelligence techniques to improve image quality, identify objects, and estimate depth in challenging visual environments. By combining CLAHE-based preprocessing, GAN-driven image enhancement, and deep learning models for object detection and depth estimation, the system provides clear, information-rich outputs that enhance both human interpretation and machine understanding of images.

While the project requires substantial computational resources and high-quality datasets for optimal performance, the modular design ensures maintainability, scalability, and potential for future enhancements. Overall, this project highlights the effectiveness of AI-driven solutions in solving complex image processing problems and provides a robust foundation for further research and real-world deployment.

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